

LearnHub: Your Center for Skill Enhancement

Project Report

1.INTRODUCTION

1.1 Project Overview

LearnHub is an online learning platform designed to simplify skill development and course management. It allows students to enroll in courses, track progress, and earn certificates. Instructors can create, manage, and deliver course content effectively. The platform features secure authentication, role-based access, and cloud file handling. LearnHub ensures scalability, user-friendly design, and personalized learning experiences.

1.2 Purpose

The purpose of LearnHub is to provide a structured, accessible, and secure platform for online learning, where students can enhance their skills and instructors can effectively create and manage courses. The goal is to make education engaging, flexible, and scalable for a wide range of users.

2.IDEATION PHASE

2.1 Problem Statement

Many learners struggle to find a structured and interactive platform to enhance their skills, while instructors face challenges in managing and delivering content effectively. Existing platforms often lack simplicity, role-based features, and personalized progress tracking, making it hard for users to engage in a seamless learning experience.

2.2 Empathy Map Canvas

The Empathy Map for LearnHub highlights the needs and experiences of a typical student user. Learners often think about finding a structured, easy-to-use platform that helps them gain real skills and track their progress.

They feel frustrated with complex or cluttered systems and are motivated by visible achievements like certificates. They observe that many platforms lack consistency or personal guidance and often hear from peers and social media about the importance of upskilling. Students say they want a simple, distraction-free environment and tend to drop out of courses that lack engagement or clear progress paths.

Example: LearnHub

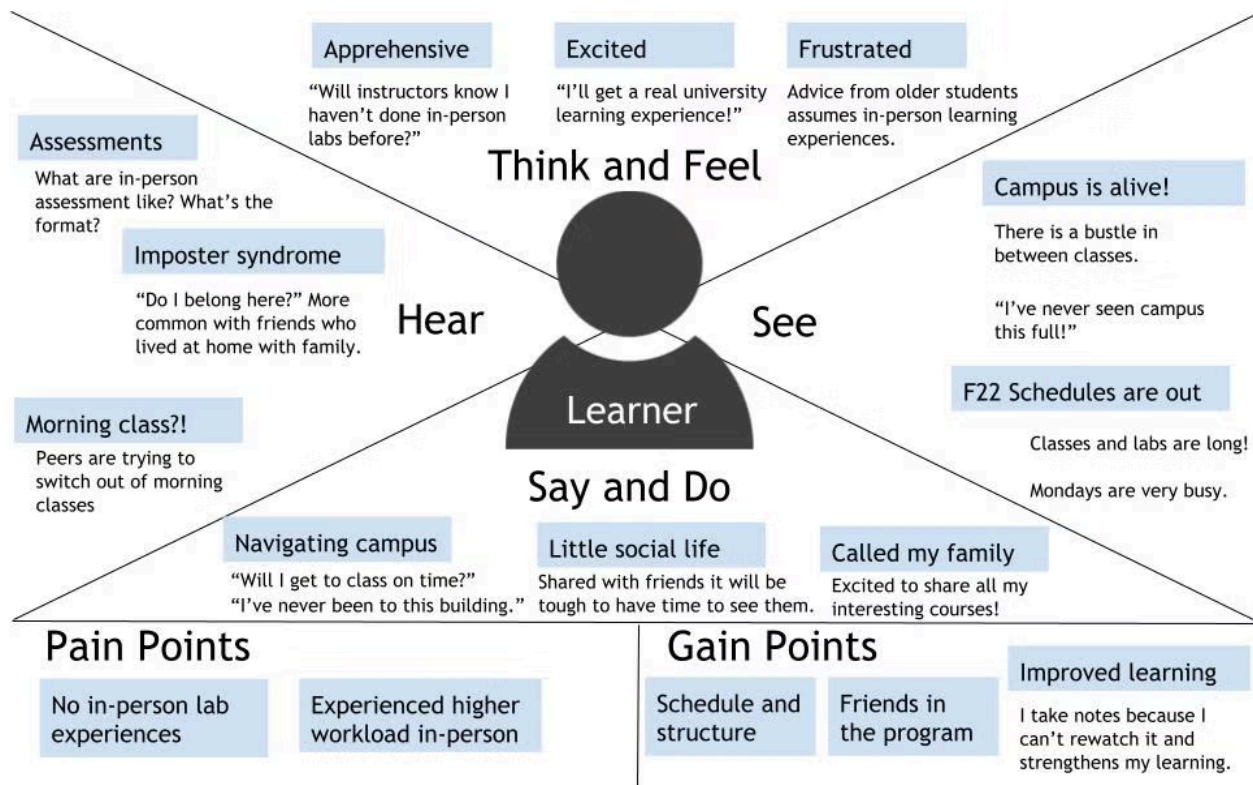


Fig: Empathy Map Canvas for LearnHub

2.3 Brainstorming

LearnHub focused on identifying the core challenges in existing online learning platforms and generating innovative solutions to address them. Key problems included a lack of structured learning paths, poor progress tracking, and limited tools for instructors to manage content effectively. To overcome these, ideas such as role-based dashboards, course creation with multimedia content, quizzes, automated certificate generation, and real-time interaction features were proposed. Technically, the platform would use MongoDB for scalable data storage, Cloudinary or AWS S3 for media uploads, and secure JWT-based authentication.

An online learning platform(OLP) is a digital platform that provides a variety of tools and resources to facilitate learning and education over the internet. These platforms have become increasingly popular, especially in recent years, as they offer flexibility and accessibility for learners of all ages and backgrounds. Here are some key features and a description of an online learning platform:

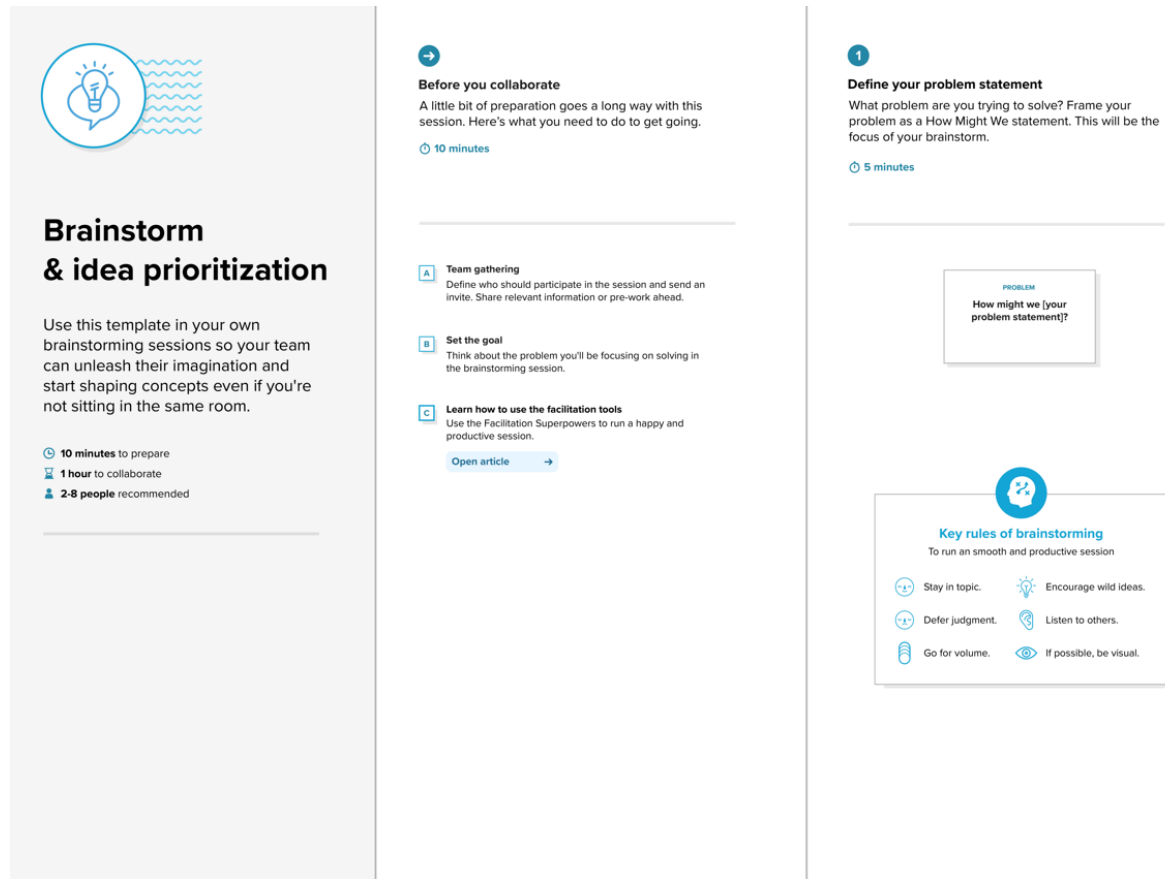


Fig: Brainstorming

- Additional enhancements like live class integration, AI-based course recommendations, gamification elements, and notification systems were also considered to boost engagement and personalization.
- The overall goal is to create a secure, user-friendly, and feature-rich learning environment that benefits both students and instructors.
- On the technical side, ideas included using a MERN stack (MongoDB, Express, React, Node.js), JWT for secure login, cloud storage for handling large media, and CI/CD pipelines for streamlined deployment.
- Future enhancements were also brainstormed, such as real-time chat between learners and instructors, integration of video conferencing tools for live sessions, AI-based skill path suggestions, gamification through badges and leaderboards, and multilingual support to reach a wider audience.
- The vision is to create a scalable, interactive, and student-centered platform that empowers both learners and educators through technology.

3.REQUIREMENT ANALYSIS

3.1 Customer Journey Map

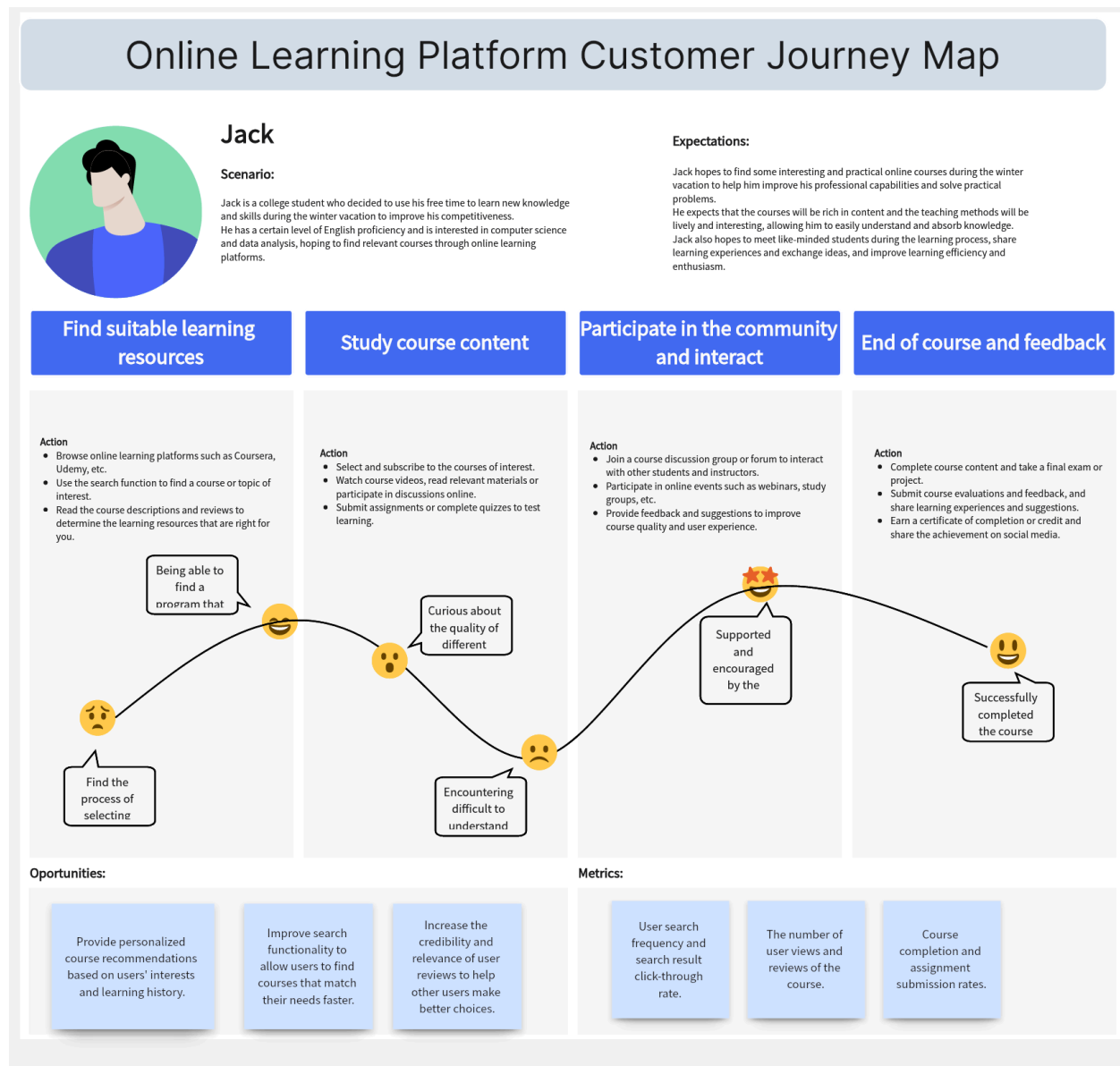


Fig:Customer Journey Map for LearnHub

3.2 Solution Requirements

Functional Requirements-Learnhub

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	Sign up,Login,Password Reset OAuth login using Google/Github
FR-2	Course Creation	Instructors can create new courses.
FR-3	Application & Enrollment	Student will log course,Learn a course

Non-functional Requirements:

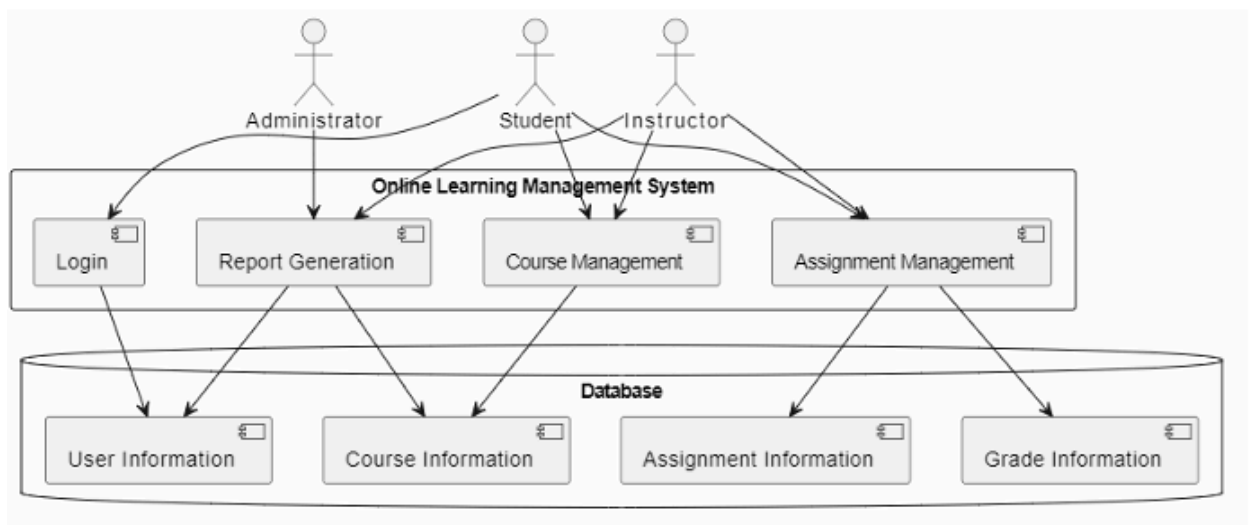
NFR No.	Non-functional Requirement	Description
NFR-1	Usability	The platform provides a simple and effective platform for users to learn, teach, and track skill development.
NFR-2	Security	It ensures secure user data and access through encrypted authentication, role-based permissions, and protected content delivery.
NFR-3	Performance	This delivers high performance with fast content loading, smooth user experience, and scalable infrastructure to support growing users.

NFR-4	Availability	That ensures high availability with reliable uptime, cloud-based hosting, and continuous access to learning resources anytime, anywhere.
NFR-5	Scalability	Built to scale efficiently, supporting increasing users, courses, and features without compromising performance or reliability.

3.3 Data Flow Diagram(DFD)

- Level 0: User ->Login/Register -> Dashboard ->Register Form ->Progress Tracking
 - Level 1: Admin -> Manage Course List -> Assign Course ->Update Course Progress
- Data Flow Diagram of LearnHub

FFi



Zero Level DFD of Online learning management system

Fig: Data Flow Diagram of LearnHub

3.4 Technology Stack

Frontend: React,Bootstrap,Material UI

Backend: Node.js, Express.js

Database: Mongoose, MongoDB

Authentication: JSON Web Tokens (JWT)

Tools: Postman, GitHub, VSCode

4.PROJECT DESIGN

4.1 Problem-Solution Fit

Many learners struggle to find a structured, engaging, and easy-to-use platform for developing skills, while instructors face challenges in creating and managing content efficiently. Most existing platforms either lack personalization, are too complex, or fail to track student progress effectively.

Purpose:

- To provide a structured and accessible platform for online learning.
- To help students learn new skills, track progress, and earn certifications.
- To enable instructors to create, manage, and deliver courses easily.
- To ensure a secure, scalable, and user-friendly digital learning experience.

Problem Statement:

- Learners face difficulty finding a structured and user-friendly platform for skill development.
- Existing platforms often lack personalized progress tracking and certification.
- Instructors struggle with complex tools for course creation and content management.
- There is a need for a secure, scalable, and interactive learning environment for all users.

Solution:

LearnHub, a full-stack online learning system, which offers:

- Provide a structured and intuitive platform for students to learn and track progress.
- Offer role-based dashboards for students, instructors, and admins with tailored features.
- Enable instructors to easily create, manage, and deliver multimedia-rich courses.
- Ensure secure authentication, cloud storage, and scalable performance for all users.

4.2 Proposed Solution:

Proposed Solution for LearnHub is as follows:

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Learners struggle to access structured, flexible online education with progress tracking and certification. Instructors face difficulties in managing and delivering engaging course content effectively.
2.	Idea / Solution description	LearnHub provides a user-friendly platform where learners can take structured courses, track progress, and earn certificates. Instructors can easily create, manage, and deliver content to engage students effectively.
3.	Novelty / Uniqueness	1.Different views for students, teachers, and admins. 2.Tracks learning progress step by step. 3.Chat, live classes, and instant certificates.
4.	Social Impact / Customer Satisfaction	1.Accessible Learning for all, anytime and anywhere. 2.Easy Course Experience with smooth navigation and progress tracking. 3.Empowers Instructors to teach and reach more students online.
5.	Business Model (Revenue Model)	Free access for users.
6.	Scalability of the Solution	LearnHub can easily grow to handle more users, courses, and features without slowing down.

4.3 Solution Architecture

The solution architecture of LearnHub is designed to be scalable, secure, and user-friendly, enabling a seamless online learning experience. The frontend is built using React.js, offering a responsive, role-based interface for students, instructors, and admins to interact with course content and track progress.

- The platform is deployed using **Vercel** for the frontend and **Render or Railway** for the backend and database, with CI/CD pipelines supporting automated deployment and updates.
- This architecture ensures LearnHub is robust, modular, and ready to scale with user demand.
- **MongoDB** with **Mongoose** is used as the database to store structured data like users, courses, progress, and feedback in a flexible and scalable way.

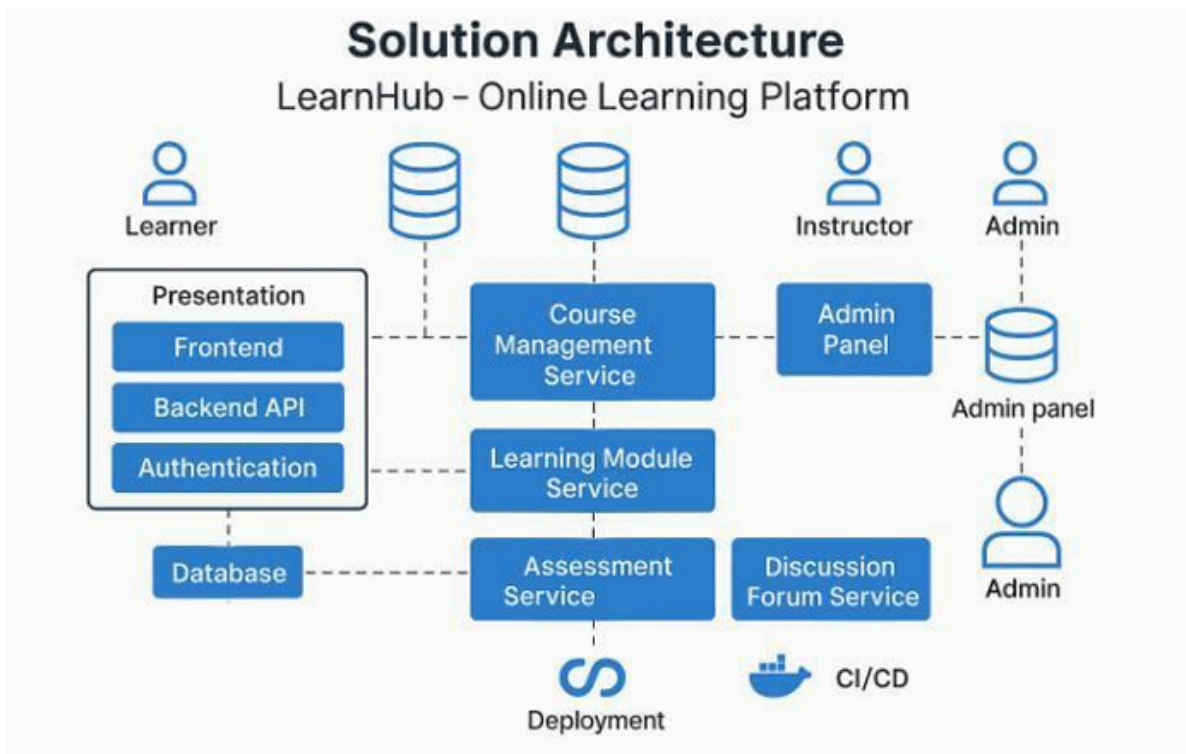


Fig: Solution Architecture for LearnHub

5.PROJECT PLANNING & SCHEDULING

5.1 Project Phases

- Requirement Gathering & Analysis
- Planning & Design
- Frontend & Backend Development
- Cloud & Media Integration
- Database Integration & Deployment
- Feedback & Enhancement

Product Backlog & Sprint Schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	User Registration	USN-1	As a User,I can sign up and login securely.	2	High	Kondareddy Kiran Kumar
Sprint-1		USN-2	As a User, I can reset my Password.	1	High	Kondareddy Kiran Kumar
Sprint-2	Course Registration	USN-3	As a user, I can handle the course list.	2	Low	Kurra Madhu Priya
Sprint-3	Application	USN-4	As an instructor,I can instruct new courses.	2	Medium	Kistaparapu Sai Durga
Sprint-3		USN-5	As a Student,I can register for a course.	1	High	Kistaparapu Sai Durga
Sprint-4	Dashboard	USN-6	As a Student,I can earn a Course Certificate.	2	Medium	Kodipilla Poojitha
Sprint-4		USN-7	As an Instructor,I can review course progress.	3	Low	Kodipilla Poojitha

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint	Total Story Points	Duration	Sprint Start Date	Sprint End Date (Planned)	Story Points Completed (as on Planned End Date)	Sprint Release Date (Actual)
Sprint-1	20	8 Days	16 June 2025	23 June 2025	20	23 June 2025
Sprint-2	20	8 Days	17 June 2025	24 June 2025	20	24 June 2025
Sprint-3	20	8 Days	18 June 2025	25 June 2025	20	25 June 2025
Sprint-4	20	8 Days	19 June 2025	26 June 2025	20	26 June 2025

6.FUNCTIONAL AND PERFORMANCE TESTING

Testing Scope:

- User registration and login
- Course form Submission
- Progress Tracking
- User Communication

Requirements to be Tested:

- **As a Student**,I should be able to sign up, log in, and access my dashboard.
- **As an Instructor**,I must be able to register and log in as an instructor.
- **As an Admin**,I should be able to manage all users (students and instructors).

Testing Environment:

- URL:<https://learnhub.example.com>

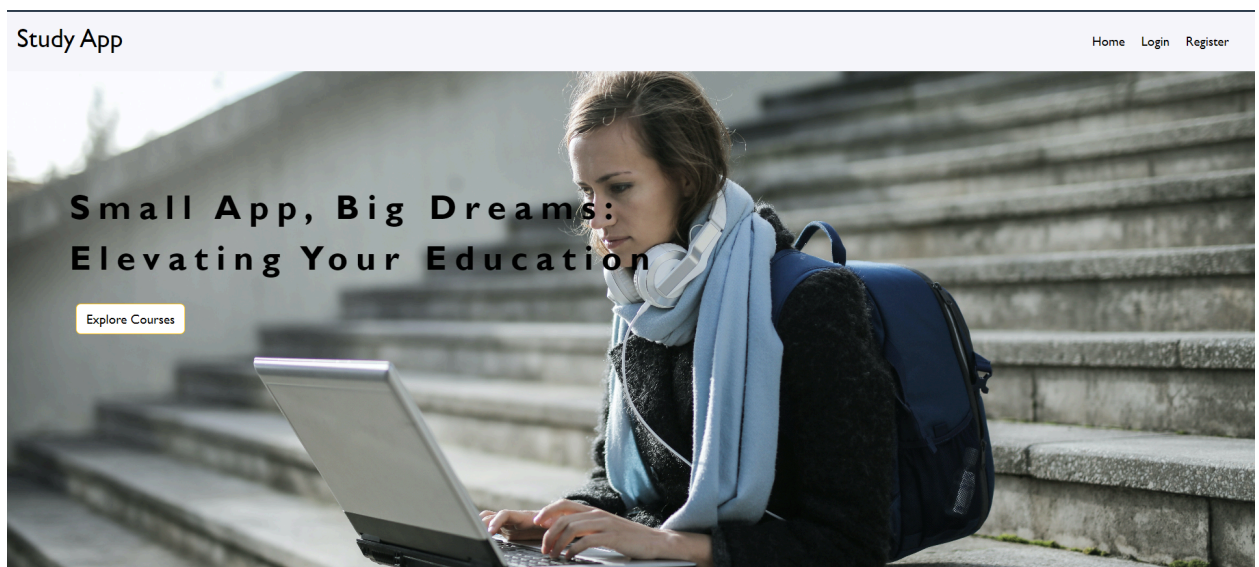
Test Cases:

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/ Fail
TC-001	User Registration	1. Visit Site 2. Click Register 3. Enter required details and submit	To be registered.	Account created, Redirected to Home.	[Pass/Fail]
TC-002	Post a job (Student)	1. Login as Student 2. Go to Dashboard 3.Enroll a Course form	To get an Enrolled course.	Enrolled courses appear in the dashboard.	[Pass/Fail]

7.RESULTS

Screenshots:

- Home Page



- Register Page

Study App

Home Login Register

 Register

Full Name

Email Address

Password

Select User

SIGN UP

Have an account? [Sign In](#)

localhost:5173/login

ENG IN 12:53 28-03-2024

- Login Page

Study App

Home Login Register

 Sign In

Email Address

Password

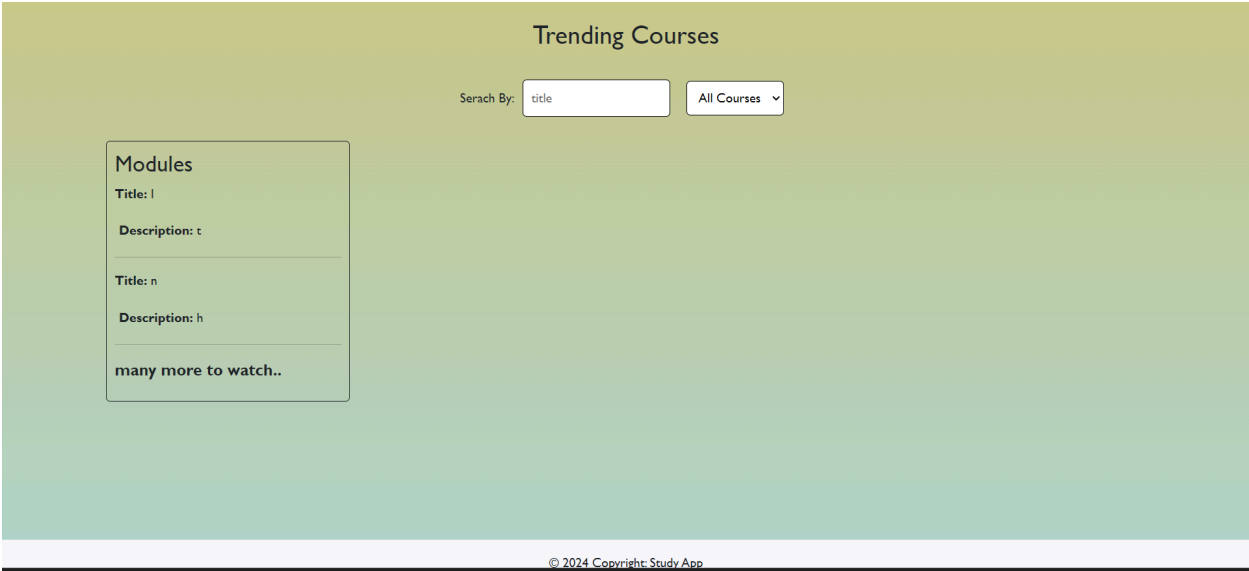
SIGN IN

Have an account? [Sign Up](#)

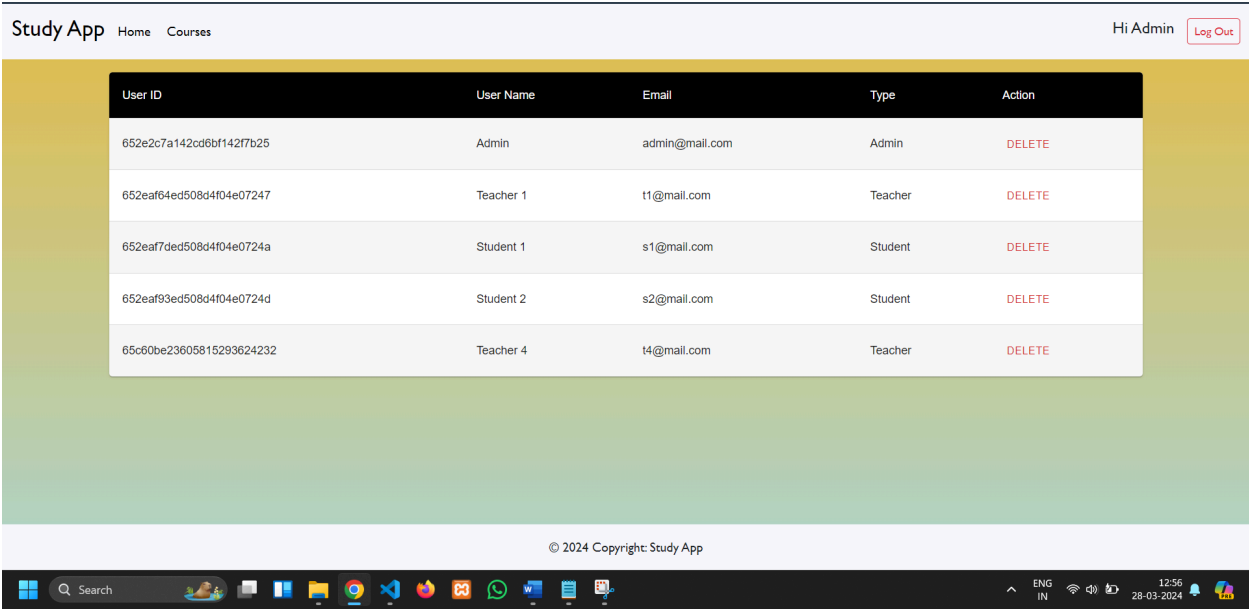
localhost:5173/login

ENG IN 12:54 28-03-2024

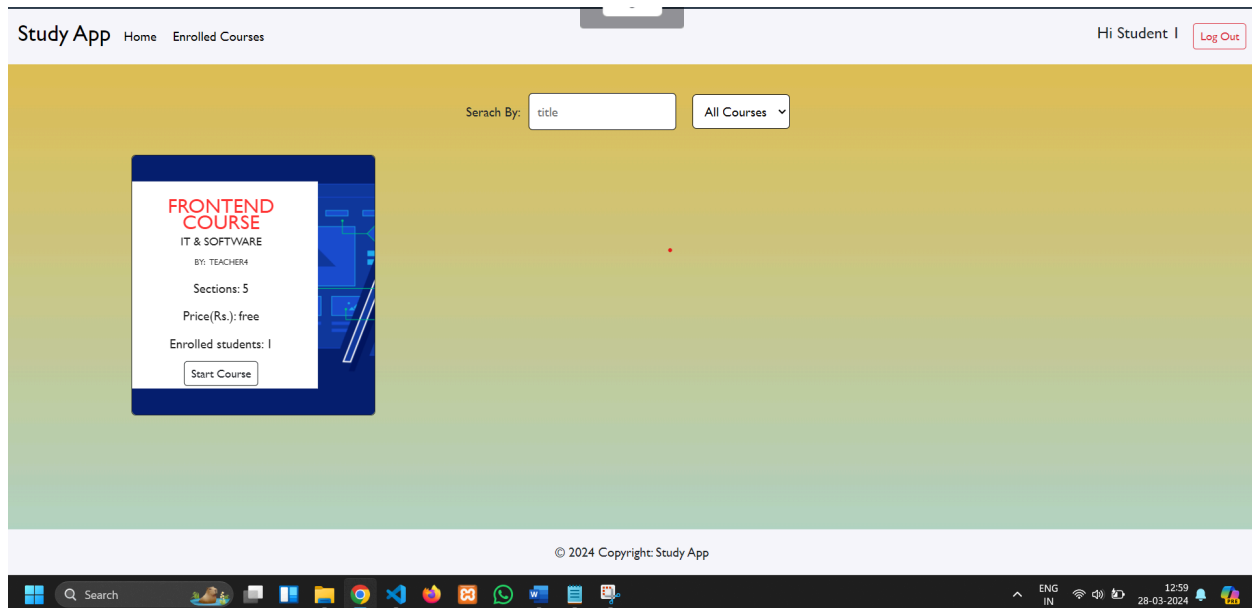
● Landing Page



● Admin Dashboard



- **Student Dashboard:**



8.ADVANTAGES & DISADVANTAGES

Advantages

- **Role-Based Access:** Different users (students, instructors, admins) get features suited to their roles.
- **Progress Tracking & Certificates:** Learners can track how far they've come and get certificates after completing courses.
- **Course Management Made Easy:** Instructors can create, manage, and upload course content without needing technical skills.
- **Flexible Learning:** Students can learn at their own pace from anywhere, anytime.

Disadvantages

- **Needs Good Internet:** Video-based content and uploads may not work well with slow internet.
- **No Personal Interaction:** Lacks real-time face-to-face interaction like in physical classrooms.
- **Dependency on Devices:** Requires a smartphone or laptop, which may not be available to everyone.
- **Security Risks (If Not Handled Well):** If authentication or data isn't secured properly, it may lead to user data leaks.

9. CONCLUSION

LearnHub is a user-friendly and flexible online learning platform designed to help students enhance their skills, instructors manage and share courses easily, and admins maintain platform quality. By offering role-based access, progress tracking, and certification, it supports a complete learning experience. While it provides many benefits like anytime learning and structured content delivery, it also requires strong internet connectivity and device access. Overall, LearnHub aims to bridge the gap between learners and educators by making digital learning more accessible, trackable, and effective.

10. FUTURE SCOPE

- Live Classes – Add real-time live sessions for better student-teacher interaction.
- Mobile App – Develop an Android/iOS app for easier access and learning on the go.
- AI Course Suggestions – Use AI to recommend courses based on user interest and activity.
- Multi-language Support – Offer content in different languages to reach a wider audience.
- Gamified Learning – Introduce badges, points, and leaderboards to make learning more engaging.

11. APPENDIX

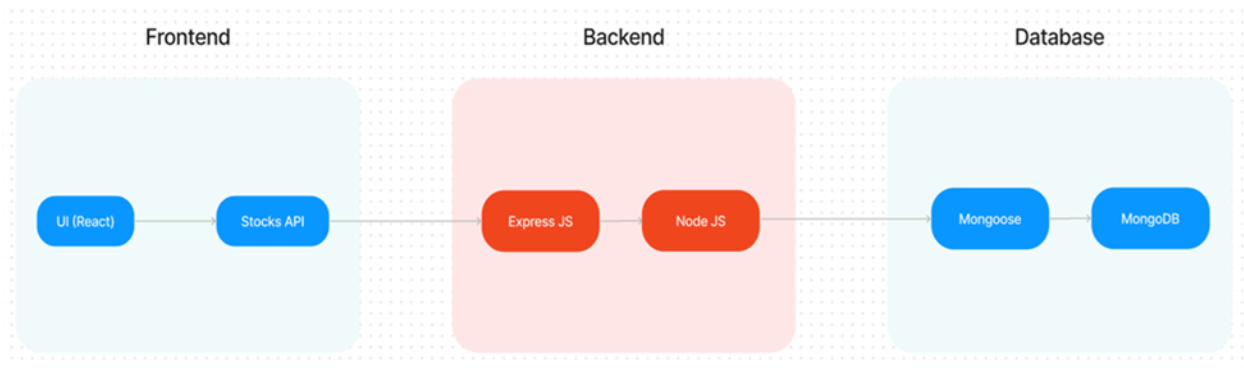
Source Code (if any):

- Complete source code is available in the GitHub repository.

GitHub & Project Demo Link:

Admin Oversight: The admin oversees the entire platform, monitoring user activity, managing course listings, and ensuring smooth operation. They keep track of enrolled students, handle any issues that arise, and maintain the integrity of the platform.

TECHNICAL ARCHITECTURE



The technical architecture of OLP app follows a client-server model, where the frontend serves as the client and the backend acts as the server. The frontend encompasses not only the user interface and presentation but also incorporates the axios library to connect with backend easily by using RESTful Apis.

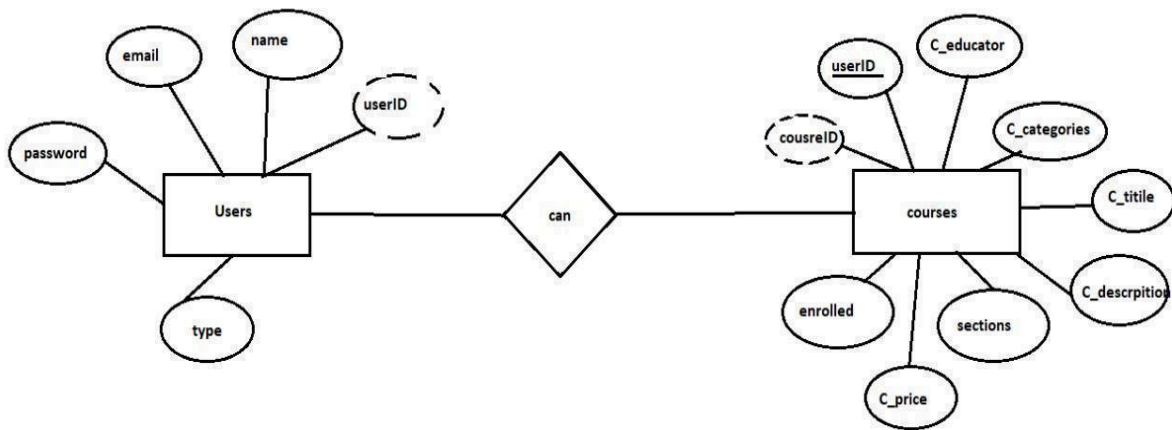
The front end utilizes the bootstrap and material UI library to establish a real-time and better UI experience for any user.

On the backend side, we employ Express.js frameworks to handle the server-side logic and communication.

For data storage and retrieval, our backend relies on MongoDB. MongoDB allows for efficient and scalable storage of user data and necessary information about the place.

Together, the frontend and backend components, along with Express.js, and MongoDB, form a comprehensive technical architecture for our OLP app. This architecture enables real-time communication, efficient data exchange, and seamless integration, ensuring a smooth and immersive blogging experience for all users.

ER Diagram



Here there are 2 collections namely users, courses that have their own fields in

Users:

1. _id: (MongoDB creates by unique default)
2. name
3. email
4. password
5. type

Courses:

1. userID: (can act as a foreign key)
2. _id: (MongoDB creates by unique default)
3. C_educator
4. C_categories
5. C_title
6. C_description
7. sections
8. C_price
9. enrolled

PRE-REQUISITES:

Here are the key prerequisites for developing a full-stack application using Node.js, Express.js, MongoDB, and React.js:

✓Vite:

Vite is a new frontend build tool that aims to improve the developer experience for development with the local machine, and for the build of optimized assets for production (go live). Vite (or ViteJS) includes a development server with ES _native_ support and Hot Module Replacement; a build command based on rollup.

```
npm create vite@latest
```

✓Node.js and npm:

Node.js is a powerful JavaScript runtime environment that allows you to run JavaScript code on the server side. It provides a scalable and efficient platform for building network applications.

Install Node.js and npm on your development machine, as they are required to run JavaScript on the server side.

Download: <https://nodejs.org/en/download/>

Installation instructions: <https://nodejs.org/en/download/package-manager/>

```
npm init
```

✓Express.js:

Express.js is a fast and minimalist web application framework for Node.js. It simplifies the process of creating robust APIs and web applications, offering features like routing, middleware support, and modular architecture.

Install Express.js, a web application framework for Node.js, which handles server-side routing, middleware, and API development.

Installation: Open your command prompt or terminal and run the following command:

```
npm install express
```

✓ MongoDB:

MongoDB is a flexible and scalable NoSQL database that stores data in a JSON-like format. It provides high performance, horizontal scalability, and seamless integration with Node.js, making it ideal for handling large amounts of structured and unstructured data.

Set up a MongoDB database to store your application's data.

Download: <https://www.mongodb.com/try/download/community>

Installation instructions: <https://docs.mongodb.com/manual/installation/>

✓ React.js:

React.js is a popular JavaScript library for building user interfaces. It enables developers to create interactive and reusable UI components, making it easier to build dynamic and responsive web applications.

Install React.js, a JavaScript library for building user interfaces.

Follow the installation guide: <https://reactjs.org/docs/create-a-new-react-app.html>

✓ **HTML, CSS, and JavaScript:** Basic knowledge of HTML for creating the structure of your app, CSS for styling, and JavaScript for client-side interactivity is essential.

✓ **Database Connectivity:** Use a MongoDB driver or an Object-Document Mapping (ODM) library like Mongoose to connect your Node.js server with the MongoDB database and perform CRUD (Create, Read, Update, Delete) operations. To Connect the Database with Node JS go through the below provided link:

<https://www.section.io/engineering-education/nodejs-mongoosejs-mongodb/>

Install Dependencies:

- Navigate into the cloned repository directory:
`cd containment-zone`
- Install the required dependencies by running the following commands:
`cd frontend`
`npm install`
`cd ../backend`
`npm install`

Start the Development Server:

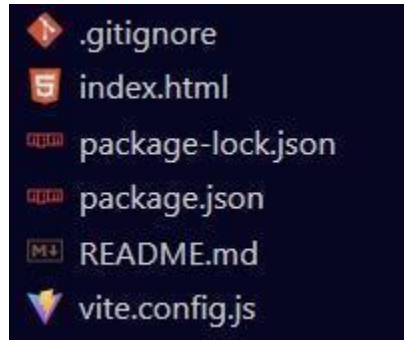
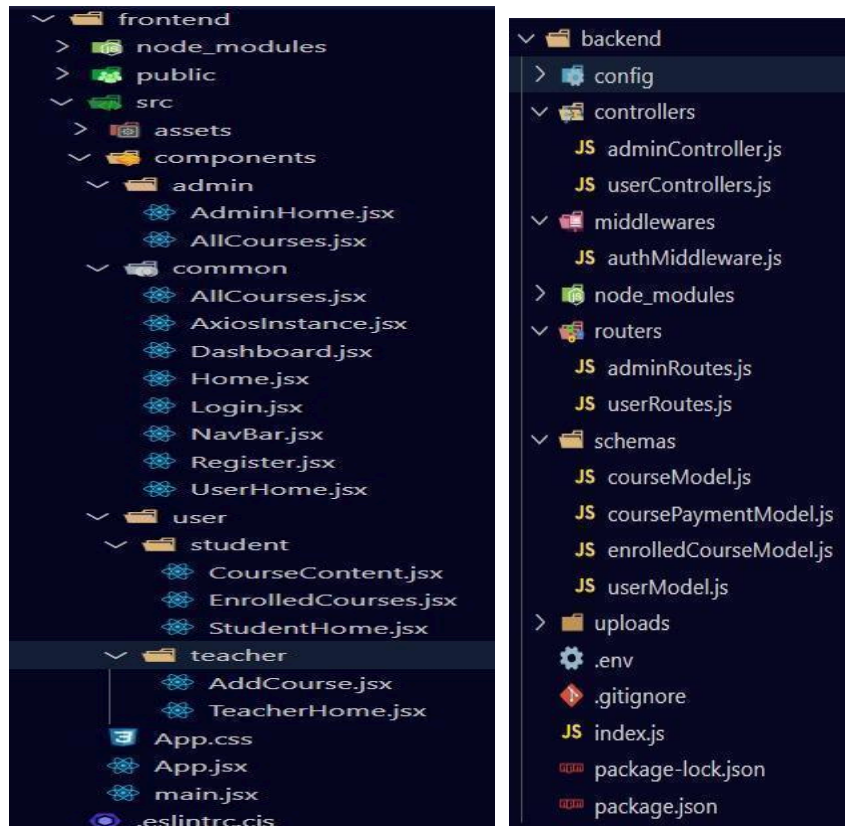
- To start the development server, execute the following command:
`npm start`
- The OLP app will be accessible at <http://localhost:5172>

You have successfully installed and set up the Online learning app on your local machine. You can now proceed with further customization, development, and testing as needed.

PROJECT STRUCTURE

The first image is of the front part which shows all the files and folders that have been used in UI development

The second image is of the Backend part which shows all the files and folders that have been used in the backend development



Application Flow: The project has a user called— teacher and student and the other will be Admin which takes care of all the users. The roles and responsibilities of these users can be inferred from the API endpoints defined in the code. Here is a summary:

Teacher:

1. Can add courses for the student.
2. Also, delete the course if no student enrolled in it or for any other reasons.
3. Also, add sections to courses.

Student:

4. Can enroll in an individual or multiple courses.
5. Can start the course where it has stopped.
6. Once the course is completed, they can download their certificate of completion of the course.
7. For a paid course, they need to purchase it and then they can start the course.
8. They can filter out the course by searching by name, category, etc

Admin:

9. They can alter all the courses that are present in the app.
10. Watch out for all kinds of users in the app.
11. Record all the enrolled students that are enrolled in the course.

Milestone 1- Setup & configuration

● Folder setup:

1. Create frontend and
2. Backend folders

. Open the backend folder to install the necessary tools

For backend, we use:

- cors
- bcryptjs
- express
- dotenv
- mongoose
- Multer
- Nodemon
- jsonwebtoken

```

{
  "name": "backend",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  ▶ Debug
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "nodemon index"
  },
  "dependencies": {
    "bcryptjs": "^2.4.3",
    "cors": "^2.8.5",
    "dotenv": "^16.3.1",
    "express": "^4.18.2",
    "jsonwebtoken": "^9.0.2",
    "mongoose": "^7.5.2",
    "multer": "^1.4.5-lts.1",
    "nodemon": "^3.0.1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

```

Milestone 2- Backend Development

- **Setup express server**

1. Create index.js file in the server (backend folder).
2. define the port number, MongoDB connection string, and JWT key in the env file to access it.
3. Configure the server by adding cors, and body-parser.

- **Add authentication:** for this,

1. You need to make a middleware folder and in that make authMiddleware.js file for the authentication of the projects and can use in.

Ref: [backend.mp4](#)

Milestone 3- Database

- **Configure MongoDB**

1. Import mongoose.
2. Add database connection from config.js file present in the config folder
3. Create a model folder to store all the DB schemas.

ref: [database.mp4](#)

Milestone 4- Frontend Development

- **Installation of required tools:**

- For frontend, we use:

1. React
2. Bootstrap
3. Material UI
4. Axios
5. Antd
6. mdb-react-ui-kit
7. react-bootstrap

```
{
  "name": "frontend",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "lint": "eslint . --ext js,jsx --report-unused-disable-directives --max-warnings 0",
    "preview": "vite preview"
  },
  "dependencies": {
    "@emotion/react": "^11.11.1",
    "@emotion/styled": "^11.11.0",
    "@mui/icons-material": "^5.14.9",
    "@mui/material": "^5.14.9",
    "axios": "^1.5.0",
    "bootstrap": "^5.3.2",
    "html2canvas": "^1.4.1",
    "jspdf": "^2.5.1",
    "mdb-react-ui-kit": "^6.1.0",
    "mdb-ui-kit": "^6.4.0",
    "react": "^18.2.0",
    "react-bootstrap": "^2.8.0",
    "react-dom": "^18.2.0",
    "react-player": "^2.13.0",
    "react-router-dom": "^6.16.0"
  },
  "devDependencies": {
    "@types/react": "^18.2.15",
    "@types/react-dom": "^18.2.7",
    "@vitejs/plugin-react": "^4.0.3",
    "eslint": "^8.45.0",
    "eslint-plugin-react": "^7.32.2",
    "eslint-plugin-react-hooks": "^4.6.0",
    "eslint-plugin-react-refresh": "^0.4.3",
    "vite": "^4.4.5"
  }
}
```

ref: [frontend.mp4](#)

Milestone 5: Project Implementation:

On completing the development part, we then ran the application one last time to verify all the functionalities and look for any bugs in it. The user interface of the application looks a bit like the one's provided below.

Landing page

Register page:

Login page:

Admin Dashboard:

Teacher Dashboard:

Course Type

Select categories

Course Title

Enter Course Title

Course Educator

Enter Course Educator

Course Price(Rs.)

for free course, enter 0

Course Description

Enter Course description

+ Add Section

Submit

Student Dashboard:

Study App

Home

Enrolled Courses

Hi Student | [Log Out](#)

Welcome to the course: Frontend Course

I

t

n

t

r

o

DOWNLOAD CERTIFICATE

QuizApp

PLAY VIDEO

0:00 / 4:45

© 2024 Copyright: Study App

ENG

IN

13:00

28-03-2024

Note: For the code drive, click on [link](#), and for the demo link, click on [project-implementation.mp4](#)